

Name : W.B.M.N.Dasuni

Reg No : DEH/IT/2324/F/070

Course Name : Programming Individual Project

Course Code : HNDIT 4052

Software Requirements Specification (SRS)

AI-Powered Internship and Job Matching Platform

1. Introduction

1.1 Purpose

This document describes the Software Requirements Specification (SRS) for the AI-Powered Internship and Job Matching Platform.

The purpose of this document is to define the functional and non-functional requirements of the system, the system interfaces, and overall system behavior.

This document will be used by developers, project supervisors, and stakeholders to understand the system requirements.

1.2 Scope

The AI-Powered Internship and Job Matching Platform is a web-based system designed to connect students, graduates, and job seekers with companies offering internships and job opportunities.

The system will:

- Allow students to create profiles and upload CVs
- Allow companies to post internships and job opportunities
- Use AI-based matching to recommend suitable opportunities
- Allow companies to review applicants
- Provide dashboards for both students and recruiters

1.3 Definitions, Acronyms

Term	Description
Admin	System administrator
AI	Artificial intelligence
DB	Database
UI	User interface
API	Application programming interface
CV	Curriculum vitae

2. Overall Description

2.1 Product Perspective

The AI-Powered Internship and Job Matching Platform is a new standalone web-based system designed to automate internship and job matching processes.

The system integrates:

- Student profiles
- Company job postings
- AI-based recommendation engine
- Application tracking system

2.2 Product Functions

Main functions of the system include:

- User registration and login
- Student profile management
- Upload CV and skills
- Company job posting management
- AI-based internship/job recommendation
- Application submission
- Admin management of system data
- Notification system

2.3 User Characteristics

Admin

- Basic to intermediate computer skills
- Responsible for system monitoring and management

Students / Job Seekers

- Basic computer knowledge
- Able to upload CVs and apply for jobs

Recruiters / Companies

- Intermediate computer skills
- Able to post job opportunities and review applicants

2.4 Assumptions and Dependencies

- Users must have internet access
- System requires database connectivity
- Users must create accounts to use the system
- AI recommendation model must be trained using skill data

3. External Interface Requirements

3.1 User Interfaces

The system will include the following interfaces:

- Login page
- User registration page
- Student dashboard
- Recruiter dashboard
- Job posting interface
- Job search interface
- AI recommendation page

3.2 Hardware Interfaces

The system requires:

- PC or Laptop
- Mobile device (optional)
- Internet connection

3.3 Software Interfaces

The system will interact with:

- MongoDB database
- Backend server
- AI recommendation module
- Web browser

4. Functional Requirements

FR1: User Registration

The system shall allow users to register as Student, Recruiter, or Admin.

FR2: User Login

The system shall allow registered users to log in using username and password.

FR3: Profile Management

The system shall allow students to create and update their profile information including skills, education, and experience.

FR4: CV Upload

The system shall allow students to upload their CV.

FR5: Job Posting

The system shall allow recruiters to post internship and job opportunities.

FR6: Job Search

The system shall allow students to search for internships and jobs.

FR7: AI Matching

The system shall recommend internships and jobs to students based on their skills and profile using AI algorithms.

FR8: Application Submission

The system shall allow students to apply for internships or jobs.

FR9: Application Management

The system shall allow recruiters to view and manage job applications.

FR10: Admin Management

The system shall allow the admin to manage users, job posts, and system data.

5. Non-Functional Requirements**5.1 Performance**

- The system should respond to user requests within 2 seconds
- The system should support multiple users simultaneously

5.2 Security

- User passwords must be encrypted
- Only authorized users can access specific system features
- Secure login authentication must be implemented

5.3 Usability

- The system must have a simple and user-friendly interface
- Navigation should be clear and easy for users

5.4 Reliability

- The system should be available 24/7
- Data should be stored securely in the database
- System failures should not cause data loss

6. System Requirements**6.1 Hardware Requirements**

- Computer or Laptop
- Minimum 4GB RAM
- Internet connection

6.2 Software Requirements

- Operating System: Windows / Linux
- Web Browser: Chrome, Edge, Firefox
- Database: MongoDB

6.3 Tools & Technologies

- Frontend: React.js / HTML / CSS / JavaScript
- Backend: Node.js / Express.js
- Database: MongoDB
- AI Matching: Python / Machine Learning Algorithm
- IDE: VS Code

7. Use Case Diagram (UML)

Actors:

- Student
- Recruiter
- Admin

Main Use Cases:

- Register
- Login
- Manage Profile
- Upload CV
- Post Jobs
- Apply for Jobs
- View Applications
- Manage System

Use Case Descriptions

Use Case 1: User Login

Actor: Student / Recruiter / Admin

Description: User logs into the system using username and password.

Precondition: User must be registered.

Postcondition: User is redirected to the dashboard.

Use Case 2: Apply for Job

Actor: Student

Description: Student selects a job and submits an application.

Precondition: Student must be logged in.

Postcondition: Application is saved in the system.

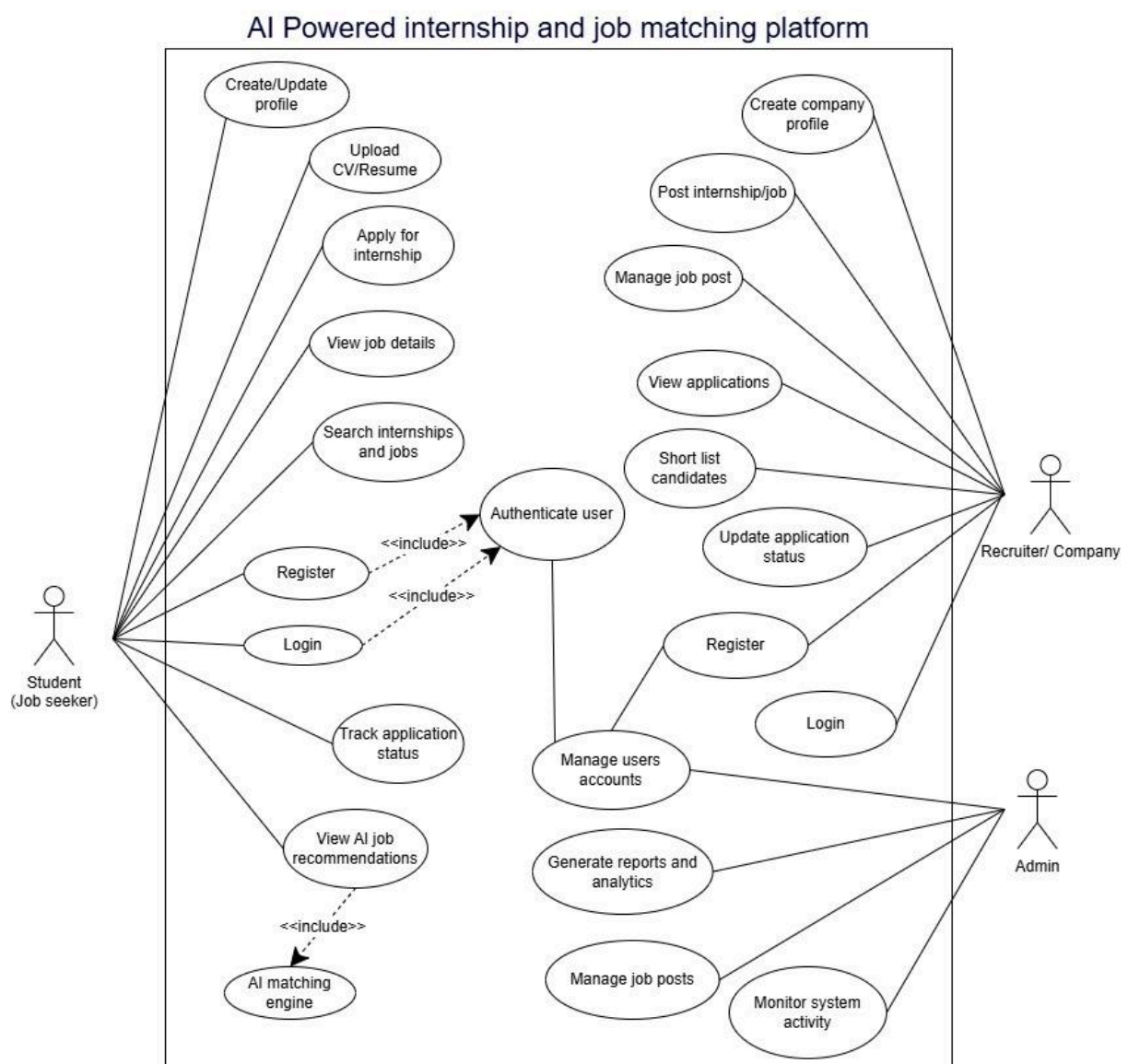
Use Case 3: Post Job

Actor: Recruiter

Description: Recruiter creates a job posting with job details.

Precondition: Recruiter must be logged in.

Postcondition: Job post becomes visible to students.



Other Requirements

Constraints

- System requires stable internet
- Users must register before accessing services

Future Enhancements

- Mobile application
- Advanced AI recommendation
- Integration with LinkedIn
- Automated resume analysis

*******END*******